

Math 142, F19
Quiz 8

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (2 points) Find the sum of the geometric series $\sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n}$.

$$a_1 = \frac{1}{4}, a_2 = -\frac{3}{16} \Rightarrow a = \frac{1}{4}, r = \frac{a_2}{a_1} = -\frac{3}{16} \times \frac{4}{1} = -\frac{3}{4}$$

$$\text{Thus } \sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n} = \frac{a}{1-r} = \frac{\frac{1}{4}}{1-(-\frac{3}{4})}$$

$$= \frac{\frac{1}{4}}{\frac{7}{4}} = \frac{1}{7}$$

Problem 2. (5 points) Determine whether the series is convergent or divergent using **Integral Test**.

(a) $\sum_{n=1}^{+\infty} \frac{1}{(n+1)^2}$

$$a_n = \frac{1}{(n+1)^2} \Rightarrow f(x) = \frac{1}{(x+1)^2}$$

$$\int_1^{+\infty} f(x) dx = \int_1^{+\infty} \frac{1}{(x+1)^2} dx$$

$$= -\frac{1}{x+1} \Big|_1^{+\infty}$$

$$= \lim_{b \rightarrow +\infty} \left[-\frac{1}{b+1} - \left(-\frac{1}{2}\right) \right]$$

$$= 0 + \frac{1}{2} = \frac{1}{2}$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{1}{(n+1)^2}$ is convergent
by Integral Test

(b) $\sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}}$

$$a_n = \frac{1}{n\sqrt{\ln n}} \Rightarrow f(x) = \frac{1}{x\sqrt{\ln x}}$$

$$\int_2^{+\infty} \frac{1}{x\sqrt{\ln x}} dx = (2\sqrt{\ln x})_2^{+\infty}$$

$$= \lim_{b \rightarrow +\infty} (2\sqrt{\ln b} - 2\sqrt{\ln 2})$$

$$= +\infty$$

$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}}$ is divergent.
by Integral Test

Problem 3. (3 points) Show the series $\sum_{n=1}^{+\infty} \frac{\cos(n)+1}{n^3}$ is convergent using **Comparison Test**.

$$a_n = \frac{\cos(n)+1}{n^3}, b_n = \frac{2}{n^3}, 0 \leq a_n \leq b_n \text{ for all } n \text{ since}$$

$$\sum_{n=1}^{+\infty} b_n = 2 \sum_{n=1}^{+\infty} \frac{1}{n^3} \text{ is convergent} \quad 0 \leq \cos(n)+1 \leq 1+1=2$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{\cos(n)+1}{n^3}$ is convergent by Comparison Test